

Telco Customer Churn Prediction

Using Apache Spark ETL and Machine Learning Pipeline

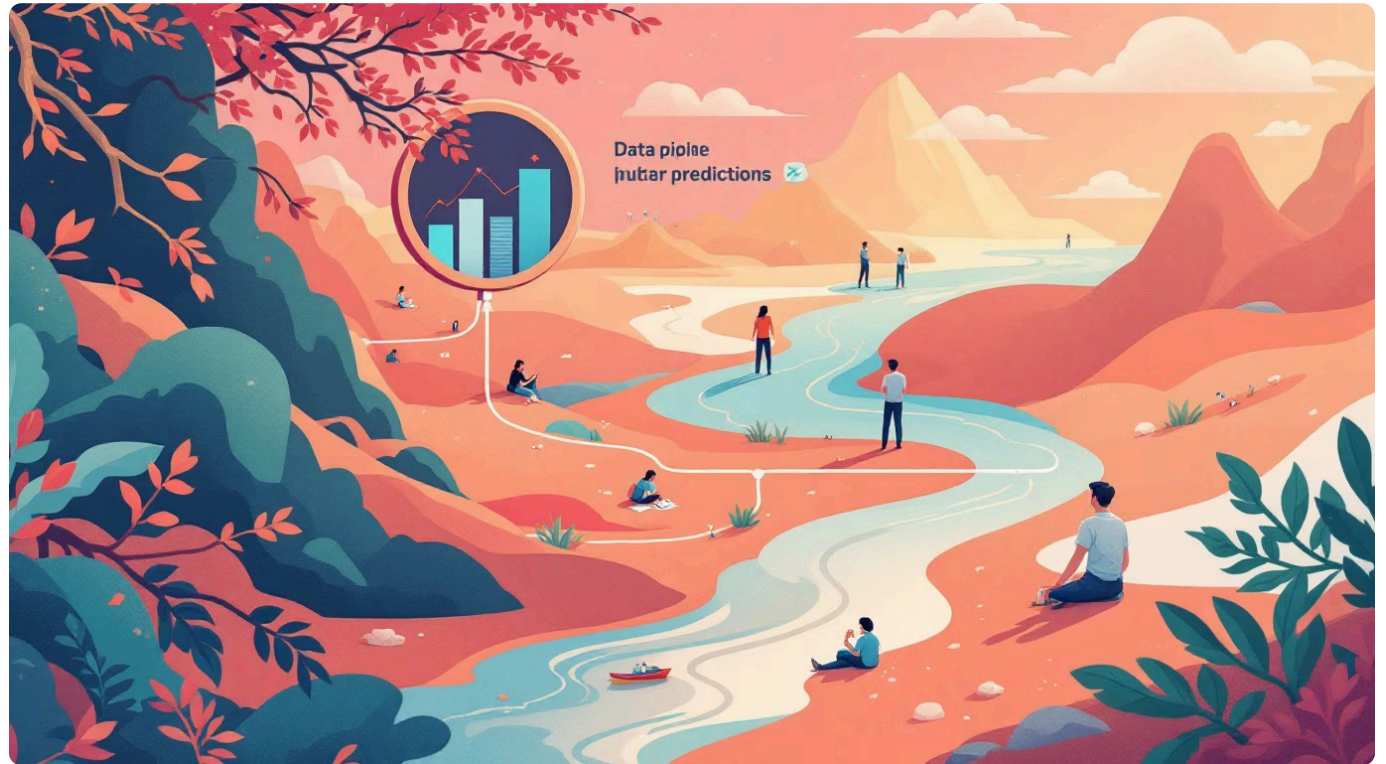
Orbin Sunny | BCR70 | BROTOTYPE | March 2026



Project Overview

Customer retention is critical in telecommunications, where churn rates range from 1.5-3% monthly. This project implements an end-to-end Big Data pipeline using Apache Spark to predict customer churn from demographic, account, and service data.

The pipeline processes 7,043 customer records with 21 features, applying ETL transformations and machine learning to achieve 80.2% accuracy in churn prediction.



Dataset Overview

7,043

Customer Records

Individual customers with complete profiles

21

Features

Demographic, account, and service attributes

26.5%

Churn Rate

Customers who cancelled services



ETL Pipeline Architecture



Extract

CSV data → Spark DataFrame with schema inference



Transform

Data cleaning, encoding, feature engineering



Load

Parquet format with Snappy compression

Data Quality Handling

- 11 null TotalCharges imputed
- String-to-numeric type casting
- Binary categorical encoding

Optimization

- In-memory caching
- Columnar storage
- Predicate pushdown

Feature Engineering



TenureGroup

Discretized tenure into lifecycle stages: New (0-12 mo), Growing (13-24 mo), Mature (25-48 mo), Loyal (49-72 mo)



AvgMonthlySpendRatio

Monthly charges vs. historical average. Ratio >1.0 signals plan upgrades or price increases affecting churn risk



Machine Learning Pipeline

VectorAssembler

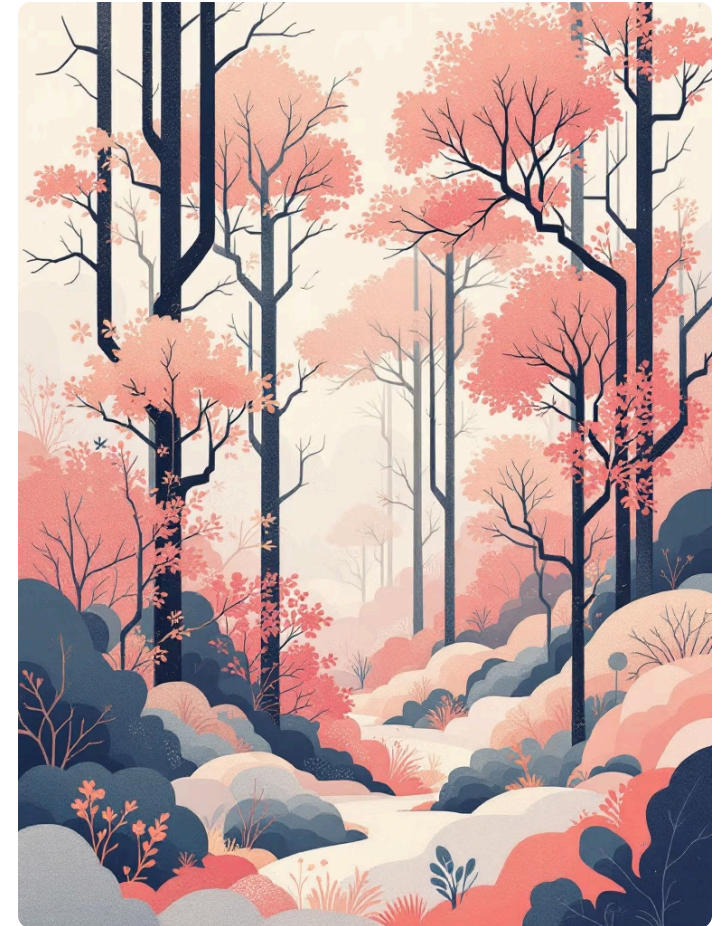
Combines 21 features into single vector for model input

Logistic Regression

Baseline linear model with elastic net regularization

Random Forest

Ensemble method with 100+ decision trees for nonlinear patterns



01

Data Splitting

80/20 train-test split with seed=42

02

Model Training

Distributed parallel processing

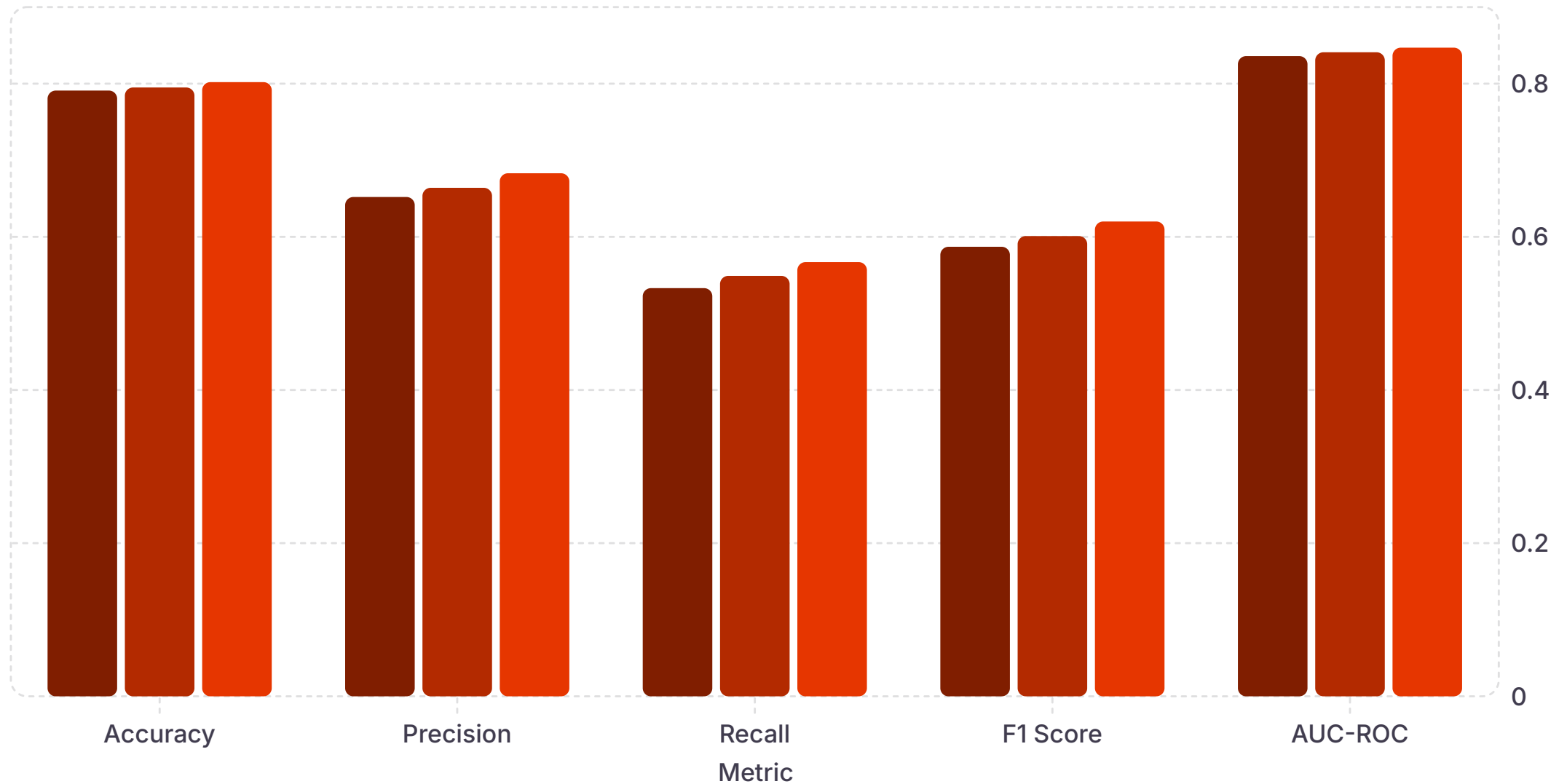
03

Evaluation

Multiple metrics on test set

Model Performance Comparison

Logistic Regression Random Forest Best (Tuned RF)



The tuned Random Forest with 200 trees and max depth 10 achieved best results, demonstrating ability to capture nonlinear feature interactions that linear models miss.

Key Findings

Feature Importance

Contract type, tenure, and monthly charges emerged as strongest churn predictors. Month-to-month contracts with fiber optic service showed highest risk.

Class Imbalance

73.5% non-churners vs. 26.5% churners required careful evaluation. Low recall (56.7%) indicates need for oversampling or threshold adjustment.

Nonlinear Patterns

Random Forest outperformed Logistic Regression by capturing interactions like contract-tenure effects that linear models cannot detect.



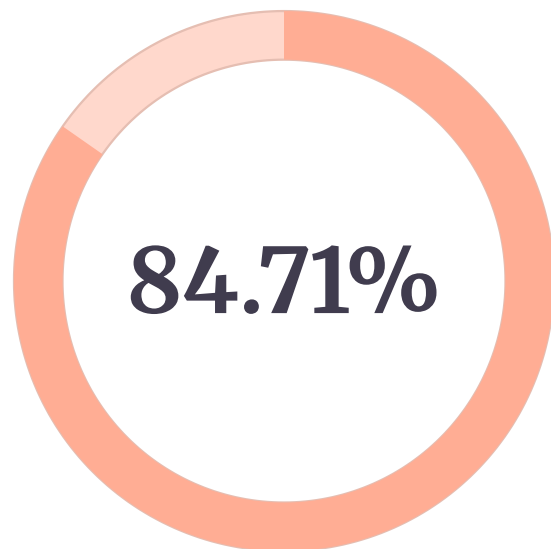
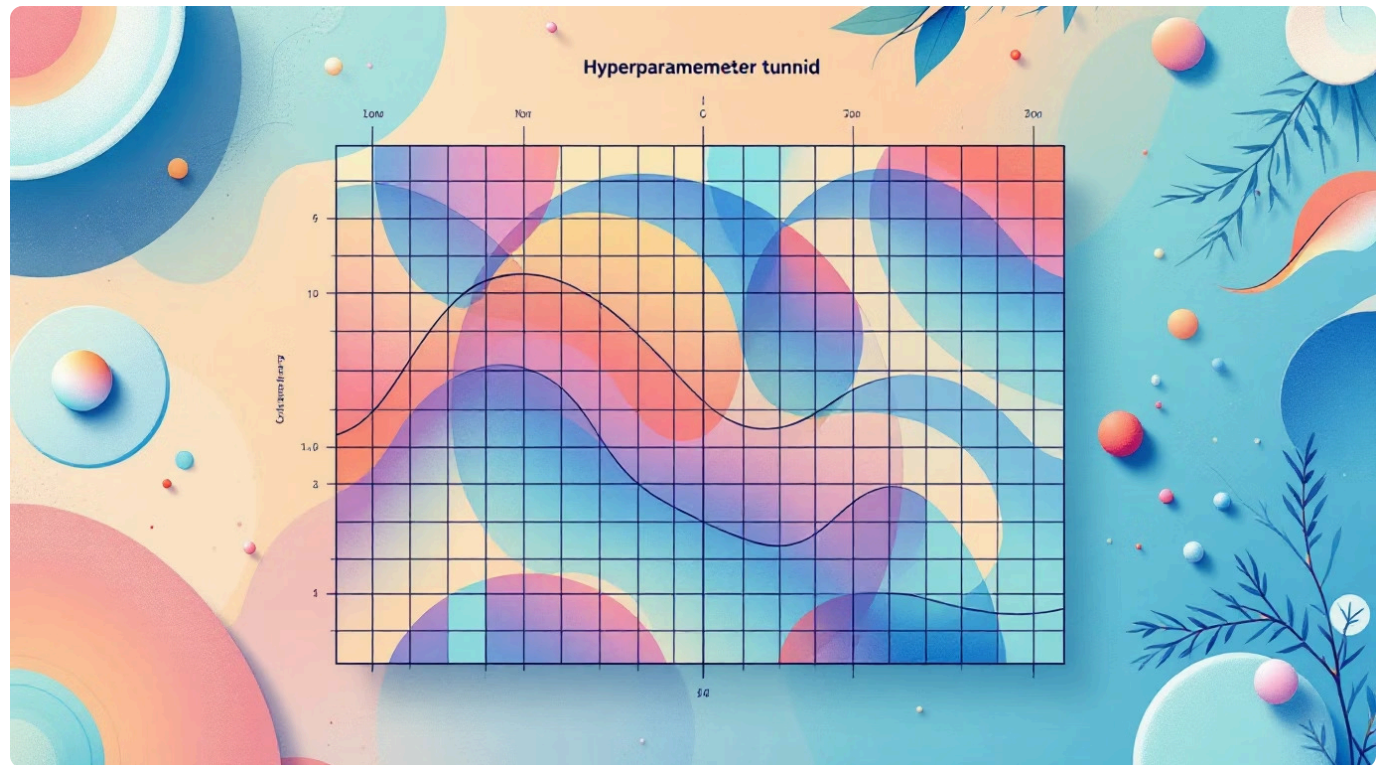
Hyperparameter Tuning

Grid Search Configuration

- Number of trees: 50, 100, 200
- Max depth: 5, 10, 15
- Feature subset: auto, all
- 18 total combinations

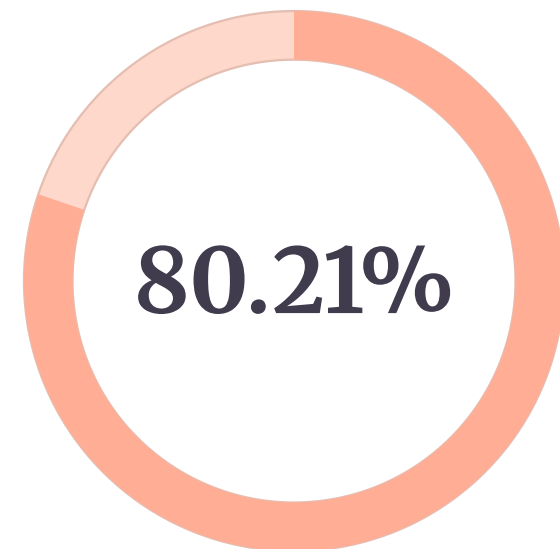
Cross-Validation

3-fold CV using AUC-ROC metric



AUC-ROC

Improved from 84.12%



Accuracy

Best overall performance



Future Enhancements



Real-Time Streaming

Spark Structured Streaming with Apache Kafka for live churn predictions



REST API Deployment

Model serving via Flask or MLflow for external applications



Interactive Dashboard

Apache Superset or Streamlit for business user visualization



Advanced Models

XGBoost, deep learning, or ensemble stacking methods